

Education

- 2025 - **PhD in Electrical Engineering and Computer Science**, *Massachusetts Institute of Technology*, USA.
- 2025 - 2026 **SM in Electrical Engineering and Computer Science**, *Massachusetts Institute of Technology*, USA.
 - Advisors: [Prof. Devavrat Shah](#) & [Prof. Stephen Bates](#).
 - Thesis Title: Causal Inference with categorical unobserved confounders via Mixture Learning
- 2023 - 2025 **Master of Statistics [M.Stat]**, *Indian Statistical Institute*, Kolkata, India.
 - Dissertation Advisors: [Prof. Aaditya Ramdas](#) (CMU) & [Prof. Soumendu Sundar Mukherjee](#) (ISI)
 - Dissertation Title: Sequential Changepoint Localization PC Mahalanobis Memorial Gold Medal
- 2020 - 2023 **Bachelor of Statistics [B.Stat] (Honours)**, *Indian Statistical Institute*, Kolkata, India.

Mrs. M.R. Iyer Memorial Gold Medal, Nikhilesh Bhattacharya Memorial Gold Medal, Usri Gangopadhyay Memorial Gold Medal

Internships / Research Visits

- Summer'24 Research Intern at the **Department of Statistics and Data Science**, *Carnegie Mellon University*.
Mentor: [Prof. Aaditya Ramdas](#)
- Summer'23 Visiting scholar at the **Big Data Summer Institute**, *School of Public Health, University of Michigan*.
- Summer'22 Summer Intern at the **Department of Mathematics**, *Indian Institute of Science*, Bangalore.
Mentor: [Prof. Manjunath Krishnapur](#)

Published (or accepted) papers/articles

In Journals

- 2026 **A. Saha**, A. Ramdas. *Post-detection inference for sequential changepoint localization*. Published in the **Journal of the Royal Statistical Society: Series B**. URL: <https://arxiv.org/abs/2502.06096>.
- 2025 **A. Saha**, A. Ramdas. *Huber-robust likelihood ratio tests for composite nulls and alternatives*. Published in the **IEEE Transactions on Information Theory**. URL: <https://ieeexplore.ieee.org/document/11283062>
- 2025 **A. Saha**, A. Ramdas. *Saha and Ramdas' discussion of "Poisson-focus" by Ward, Dilillo, Eckley and Fearnhead*. Published in the **Journal of the American Statistical Association (JASA)** URL: <https://doi.org/10.1080/01621459.2024.2402957>
- 2024 **A. Saha**, N. R. Pal. *Group-Feature Selection Using a Neural Network With Group Lasso Regularization and Controlled Redundancy*. Published in **Neurocomputing**, Volume 610, 2024, 128596, ISSN 0925-2312. URL: <https://doi.org/10.1016/j.neucom.2024.128596>

In Conference Proceedings

- 2024 **A. Saha**, A. Ramdas. *Testing Exchangeability by Pairwise Betting*. Published at the **International Conference on Artificial Intelligence and Statistics (AISTATS), 2024**. URL: <https://proceedings.mlr.press/v238/saha24b.html> **ORAL**

- 2023 J. Ray Choudhury*, **A. Saha***, S. Roy, S. Dutta. *Robust Classification of High-Dimensional Data using Data-Adaptive Energy Distance*. Published at the **European Conference on Machine Learning and Knowledge Discovery in Databases: Research Track. (ECML PKDD), 2023**. URL: https://doi.org/10.1007/978-3-031-43424-2_6

Preprints / Working Papers

- 2026 **A. Saha**, A. Ramdas. *E-detectors for non-partitioned sequential change detection*. URL: <https://arxiv.org/pdf/2607.28322>
- 2026 **A. Saha**, S. Bates, D. Shah. *Causal Inference with categorical unobserved confounders via Mixture Learning*. URL: <https://arxiv.org/abs/2605.19006>.
- 2026 **A. Saha**, A. Ramdas. *Distribution-free changepoint localization after sequential change detection*. URL: <https://arxiv.org/abs/2606.01256>.
- 2026 **A. Saha**, A. Ramdas. *Optimal e-variables under constraints*. URL: <https://arxiv.org/abs/2604.21680>.
- 2025 **A. Saha**, A. Ramdas. *Density estimation with atom and functional estimation for discrete-continuous mixture*. URL: <https://arxiv.org/abs/2508.01706>.

Academic Accolades

- 2026 **National Science Foundation (NSF) Travel Award** for participation in the 9th International Workshop in Sequential Methodologies (IWSM-2026) at American University, Washington, DC.
- 2025 Awarded **Prasanta Chandra Mahalanobis Gold Medal** as the **most outstanding** student in the M.Stat. programme (2023 - 2025) at the Indian Statistical Institute
- 2025 Awarded **Hewlett Packard Fellowship** by MIT School of Engineering
- 2025 Awarded **EDGE Doctoral Fellowship** by Stanford University [Declined]
- 2023 Received **Mrs. M.R. Iyer Memorial Gold Medal** for securing the **first position** in B.Stat (Hons.) program (2020 - 2023) at the Indian Statistical Institute
- 2023 Received **Nikhilesh Bhattacharya Memorial Gold Medal** for the **best performance in Statistics** in B.Stat (Hons.) program (2020 - 2023) at the Indian Statistical Institute
- 2023 Received **Usri Gangopadhyay Memorial Gold Medal** in B.Stat (Hons.) program (2020 - 2023) at the Indian Statistical Institute
- 2020 - 2023 Received the **prize money for outstanding performance** in all semesters in the B.Stat program (2020 - 2023) at the Indian Statistical Institute (ISI), Kolkata
- 2021 & 2022 Received **Mukul Chaudhuri Cash Award** in both 1st year and 2nd year of B.Stat program from the Indian Statistical Institute (ISI), Kolkata
- 2020 - 2025 Received the **Kishore Vaigyanik Protsahan Yojana (Young Scientist Incentive Plan) fellowship**, funded by the Department of Science and Technology, Government of India
- 2019 & 2020 Selected **twice** to participate in the **Asian Pacific Mathematical Olympiad (APMO)** from India
- 2019 & 2020 Selected **twice** to participate in the **International Mathematical Olympiad Training Camp (IMOTC)** organized by Homi Bhabha Centre for Science Education (HBCSE), TIFR

*Equal contributions

- 2019 Qualified **Indian National Mathematical Olympiad (INMO)** (among top ~75 participants all India) organized by the NBHM, Government of India.
- 2018 - 2019 Qualified aptitude test and interview rounds in SA stream of **Kishore Vaigyanik Protsahan Yojana (KVPY)** exam with an **All India Rank of 388**

Other Academic Experiences

- Jul, 2024 Participant of a summer school at **Columbia University**
- Jul, 2021 Participant of **Mathematical Summer in Paris** (online) organised by *CNRS/ Université Paris-Saclay*
- May, 2020 Participant (as a member of the senior batch) of **International Mathematical Olympiad Training Camp (IMOTC)** organized by *Homi Bhabha Centre for Science Education (HBCSE), Mumbai*
- Apr - May 2019 Participant of **International Mathematical Olympiad Training Camp (IMOTC)** at *Homi Bhabha Centre for Science Education (HBCSE), Mumbai*
- Dec 2018 - Jan 2019 Participant of **Indian National Mathematical Olympiad Training Camp (INMOTC)** at *Indian Statistical Institute, Kolkata*

Talks and Poster Presentations

Workshop/Conference/Seminar

- July, 2026 Poster presentation at the 4th Workshop on Sequential Anytime-Valid Inference (SAVI), University of Twente, the Netherlands
Title: Post-detection inference for sequential changepoint localization
- June, 2026 Talk & poster presentation at Causal Inference Workshop, held in conjunction with ACM SIGMETRICS, Ann Arbor, USA
Title: Causal Inference with Categorical Unobserved Confounder via Mixture Learning
- June, 2026 Invited talk at the International Workshop in Sequential Methodologies (IWSM), Washington DC, USA
Title: Post-detection inference for sequential changepoint localization
- July, 2025 Invited talk at the 3rd Workshop on Sequential Anytime-Valid Inference (SAVI), Chennai Mathematical Institute, India
Title: Huber-Robust Likelihood Ratio Tests for Composite Nulls and Alternatives
- May, 2025 Invited talk (online) at International Seminar on Selective Inference (ISSI)
Title: Post-detection inference for sequential changepoint localization
- Jul, 2023 Invited talk at the Symposium on Big Data, Human Health, and Statistics, University of Michigan, USA
Title: Machine Learning for Healthcare

University/Departmental Seminar

- Sep, 2025 [Invited talk] **PCM Gold Medal Presentation** at the Indian Statistical Institute, Kolkata
Title: Testing exchangeability by pairwise betting [\[slides\]](#)
- Jul, 2024 Presentation at Carnegie Mellon University, USA
Title: Huber-Robust Anytime-Valid Sequential Probability Ratio Test
- Sep, 2023 [Invited talk] **D. Basu Memorial Award Presentation** at the Indian Statistical Institute, Kolkata

Title: Navigating the Challenges of High-dimensional Low Sample Size Data: A Two-sample Test and a Novel Classification Approach [\[Link to the slides\]](#)

Industry

Feb, 2025 Invited talk (online) at Amazon Research (AWS)

Title: Post-detection inference for sequential changepoint localization

Additional Information

- **Hobbies:** Painting, Chess, Swimming, Traveling, Table tennis
 - **Service**
 - A **reviewer** of Sankhya A, Bernoulli.
 - **Co-chair** of the [31-th Annual Sanjoy K. Mitter LIDS Student Conference](#), 2026.
 - A **core member** of the organizing committee of the [Math Club](#) at ISI Kolkata, 2023-2025.
 - Served as a member of the question paper committee for the [Mathematics Talent Reward Programme](#) (MTRP, 2023), a math contest for high school students organized by ISI Kolkata students.
 - **Google Scholar:** <https://scholar.google.com/citations?user=fKv7P-wAAAAJ&hl=en>
 - **GitHub:** <https://github.com/Aytijhya>
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